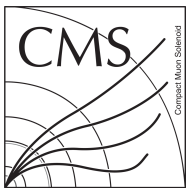
Internal review draft – 22 October 2026
No CMS DP number assigned



The Compact Muon Solenoid Experiment
CMS Performance Note



Dual-threshold time-walk calibration for CMS BTL

CMS

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Abstract

With new high intrinsic precision timing detectors for the HL-LHC upgrades, the strategy for commissioning these detectors must differ from previous collider experience. Prior to LHC, collider experiments benefitted from a clear timing reference for use in calibration of the detectors. At low luminosity an interaction trigger was useful for rejecting background but also fulfilled this role. For the ATLAS and CMS upgrades, an interaction trigger would be superfluous since they are designed for 140-200 interactions per trigger event. Dual threshold timing enables a calibration strategy when a t_0 reference is not readily available. We extend an earlier bench test demonstration of this technique with the CMS Barrel Timing Layer(BTL) sensors and the TOFHIR readout ASIC to a 180 GeV pion beam exposure. These data confirm the earlier single coefficient correction for walk and the conclusion that walk calibration to the required $\sim 10\%$ precision is practical with simple data sets using only this system with beam or cosmic rays.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

Dual-threshold time-walk calibration for CMS BTL

CMS

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Discussion draft; not approved for public release.

Abstract

This study examines a charge-dependent time-walk model for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. The current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

The comparison uses slope measurements from run 4076 and independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A fit determines a free constant offset and the correction amplitude only in charge intervals populated in both measurements. This is a comparison of calibration shapes, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory model that predicts the correction amplitude from internal pulse information. This model could enable operation without an event-by-event external time reference. Its transferability and full uncertainty remain to be established.

Two complementary runs, not the same events

Sample	Available information	Role in this study
Run 4076	Valid dual-threshold timing and charge	Measure the pulse-slope proxy $S(Q)$
Run 4399	MCP-associated reference timing and charge	Measure the empirical walk $W_{\mathrm{MCP}}(Q)$

The runs contain different events; no event-by-event matching is performed. The comparison transfers a channel-response model between runs.

The upstream and downstream modules each contain bars read out at two ends. Detector/readout context is given in Refs. [3, 7]. The channel-to-bar mapping determines the coincidence pairs; the same channel number is compared across the two runs.

The comparison assumes the same nominal detector setup in both runs, with MCP reference timing in run 4399 and dual-threshold timing in run 4076. Cross-run response differences are not included in the quoted uncertainties.

From two crossing times to a walk model

For thresholds $I_1 < I_2$ and their valid crossing times t_1, t_2, define

$$S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}], \qquad \overline{S}(Q) \simeq c_0 + c_1 Q.$$

S is a finite-interval rising-edge slope proxy. It is not the current difference ΔI itself and is not necessarily the instantaneous derivative at the first threshold.

$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

The amplitude A (AWC) has units of current and is constant for a channel and operating configuration. The correction varies event by event through Q.

Timing reconstruction and threshold scale

The run-4076 sample is file `23_e.root`. For coarse codes C_1, C_2,

$$d_c = (C_2 - C_1) \bmod 1024, \qquad t_2 = t_1 + (6250\,\mathrm{ps})d_c - f_1 + f_2.$$

Subtraction uses signed 64-bit coarse codes. The coarse-time term uses floating-point arithmetic to avoid the previous 16-bit multiplication overflow.

The fine-time lookup stores the mean of t_1 mod $6250\,\mathrm{ps}$ for each (channel, TAC, fine-code) combination, using all input hits. This lookup supplies f_1 and f_2, with entries for every selected hit in the 22 audited channels.

All selected hits have threshold register `step2=113621`, decoded as DAC codes 10 and 35. The implemented scale gives

$$\Delta I = (35 - 10) \times 0.3125\,\mu\mathrm{A} = 7.8125\,\mu\mathrm{A}, \qquad \frac{S}{\mu\mathrm{A/ns}} = 1000\,\frac{\Delta I/\mu\mathrm{A}}{(t_2 - t_1)/\mathrm{ps}}.$$

GitHub: timing and threshold reconstruction. This verifies the implemented conversion, not an independent calibration of the hardware thresholds.

Selection and profile construction

- Require one hit at each end of the target bar and no other hit in that module. The other module is unconstrained; channel 192 is ignored.
- Require finite Q, S, $\Delta I > 0$, $t_2 - t_1 > 0$ and $d_c = 0$. This selects the same-coarse-bin prompt population.
- Require $Q \geq Q_{\mathrm{left}}$, the left half-height boundary of the smoothed charge histogram. The Landau–Gaussian fit does not set this cut. There is no additional upper-charge cut or 100 ps minimum-rise cut.
- Form ten equal-width bins over the selected charge range. Plot mean S and hit SD; fit with SEM weights. Bins with undefined or negligible SEM are omitted.
- Bootstrap selected hits with fixed bin edges and refit 1000 times. Quote the 16th–84th percentiles.

All selected hits are used; there is no training or test split. GitHub: pair selection and profiling, with the newer optional minimum-rise cut disabled.

Reproducing the frozen run-4076 profiles

Reprocessing `23_e.root` reproduces the selected counts, ten-bin profiles and linear-fit coefficients for all 22 channels. The saved bootstrap coefficients reproduce the reported percentile intervals.

Channel	Target-pair hits	Prompt hits	Peak-right hits
137	25979	20789	19126

For channel 137, the histogram-derived boundary is $Q_{\mathrm{left}} = 184.9163$ ADC. Each profile point is placed at the mean measured Q in its bin, not at the geometric bin center.

The frozen selection retains rises below 100 ps in channels 130, 139, 151, 159, 236 and 238. These events remain in the sample. Their effect requires selection-sensitivity and channel-quality studies.

Channel 135 has one bin with a single entry. Its SEM is undefined, so only nine bins enter the linear fit. The audit changed neither the fits nor the plotted data.

Reference version shared with the MCP reference note [6]

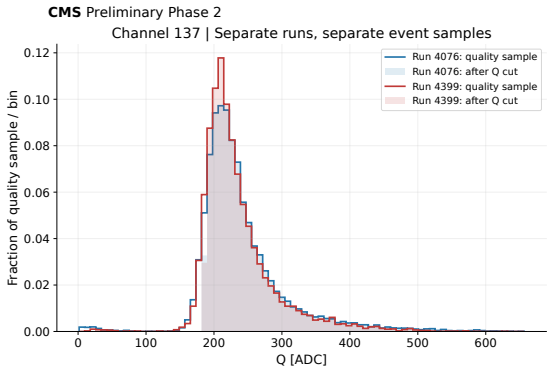
The run-4399 MCP functions were refitted on 22 October 2026. The updated procedure detects segment boundaries across missing trigger timestamps and excludes invalid clock groups. The MCP reference note documents the timing validation and remaining limitations.

For each topology, file and segment, the scaled MAD of the channel-192 versus scope-trigger line residuals must not exceed 1 ns before LOWESS [11]. The selection uses clock information, not SiPM timing residuals. Suspected incorrect event associations are excluded; MCP rows are not shifted.

All AWC-and-offset fits in this note use these updated references. The run-4076 slope fits, bootstrap replicas and measured-support matching definition are unchanged. The charge spectra use the updated reference sample.

The plotted AWC intervals hold the reference fixed. They exclude reference-fit uncertainty, clock-selection uncertainty and uncertainty in transfer between runs. The uncertainty table later in this note adds reference-fit statistics separately.

Charge spectra define the available comparison sample

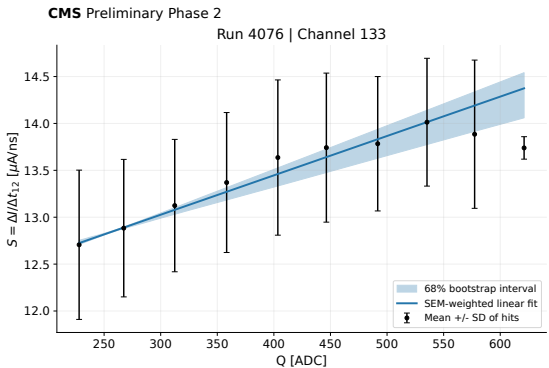


The spectra show the channel-137 charge distributions in the two independent runs. Lines show hits passing the quality selection; shaded spectra also require the charge-peak selection. Each run is normalized to its own quality-selected count.

The charge proxy is in ADC units, not MeV. The samples differ in population and acceptance, so the curve alignment uses only charge intervals populated in both runs.

Recovered histogram counts, not a new peak fit. Run-4076 profile selection starts at about 184.9 ADC for this channel.

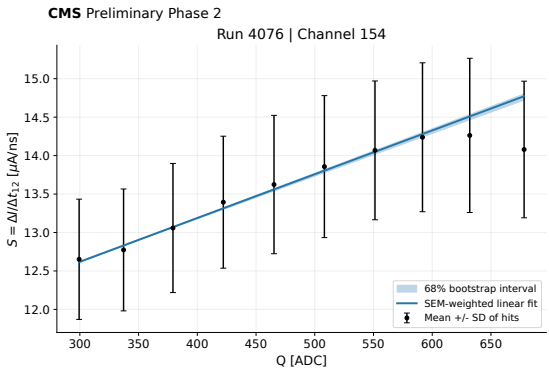
Channel 133: fitting the slope profile



Run 4076 channel-133 hits are grouped into ten charge bins. Points show mean S; bars show the SD of individual selected hits. The line is the SEM-weighted fit $c_0 + c_1 Q$.
The shaded interval is the central 68% bootstrap interval of the fitted mean curve, shown without enlargement.

SD is the hit standard deviation; SEM is the standard error of the mean. The band describes uncertainty on the fitted mean curve.

Channel 154: the opposite end of bar 7

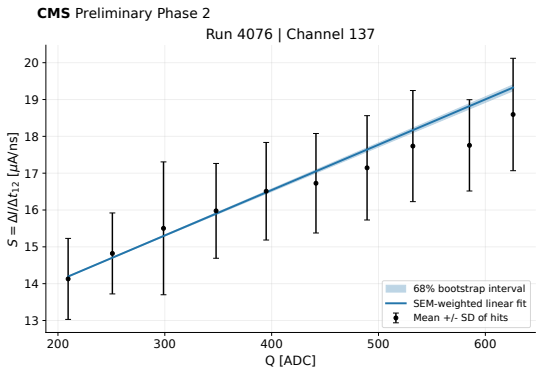


Channel 154 uses the same selection and profiling procedure. The mean slope increases with charge, but the response is calibrated independently at each bar end.

Bars show the hit SD. The line uses SEM weights. The band propagates the bootstrap coefficient replicas, including their correlation. Linearity is a working approximation, not a validated precision model. Model and selection dependence are excluded from the quoted errors; the separate sensitivity study is retained outside this note.

Run 4076, saved profile result, displayed without refitting.

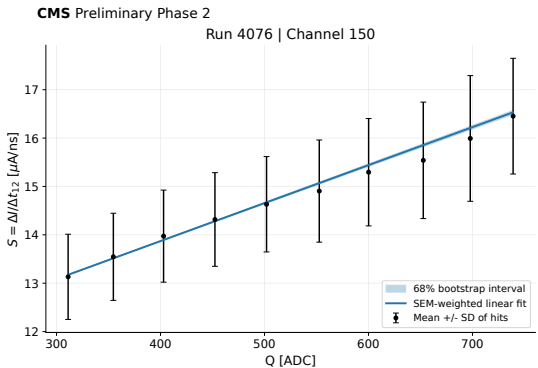
Channel 137: pulse slope and charge



The selected run-4076 sample contains 19 126 channel-137 hits. Ten profile points constrain a linear pulse-slope model over the observed charge range. The SD bars show event-to-event spread, while SEM weights determine the fit. The narrow shaded band is the conditional 68% bootstrap interval of the mean prediction, not a single-hit uncertainty.

The count and fit come from the saved profile results. Uncertainty bands are shown without enlargement.

Channel 150: the opposite end of bar 8



Channel 150 uses the same profiling procedure. The charge-dependent slope provides the denominator of the proposed walk function; a separate amplitude A determines the correction strength.

The line and shaded interval describe the fitted mean. Points and SD bars show the spread within the selected charge bins.

Run 4076, saved profile result. A constant AWC gives a charge-dependent timing correction.

Determine AWC using only jointly measured support

The run-4399 reference is a fitted quadratic in inverse charge,

$$W_{\mathrm{MCP}}(Q) = a + b/Q + c/Q^2.$$

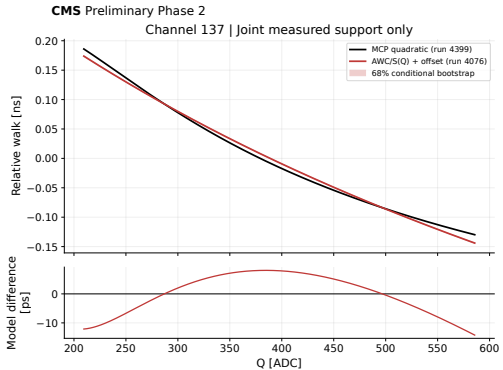
Let Ω be the intersection of the two outer profile-mean ranges, restricted to charge bins populated in both runs. Fit

$$(A, B) = \arg\min_{A,B} \sum_{Q_k \in \Omega} \left[\frac{A}{c_0 + c_1 Q_k} + B - W_{\mathrm{MCP}}(Q_k) \right]^2 .$$

The fit samples a uniform 3000-point charge grid, then removes intervals without data support. The offset B absorbs the arbitrary time zero.

This is uniform-charge shape matching, not an event-weighted likelihood. It uses neither extrapolated tails nor a single central anchor point. The uncertainty study below evaluates sensitivity to charge weighting and correlated MCP-fit uncertainty. Uniform Q is the nominal choice; it has not been established as uniquely optimal.

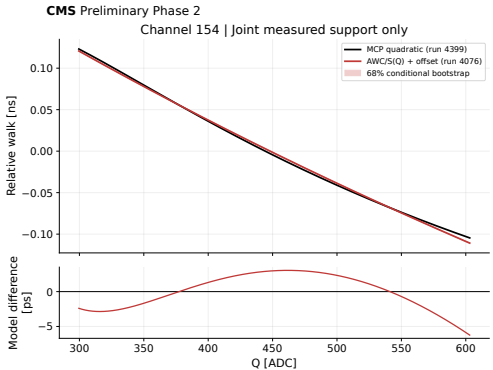
Channel 137: inverse-slope and MCP walk shapes



The black curve is the run-4399 MCP quadratic. The red curve is the run-4076 inverse-slope model after fitting AWC and a constant offset. The curves are shown only in charge intervals populated in both runs. Lower panel: run-4076 model minus MCP model. Its RMS shape mismatch is 6.80 ps on the supported uniform-Q grid, not an event timing resolution. The band varies the run-4076 slope fit while holding the MCP curve fixed. The uncertainty table later in this note also includes MCP-fit uncertainty.

AWC = 18.332 μA; conditional 16–84% interval [18.120, 18.607] μA. Both curves share one display offset.

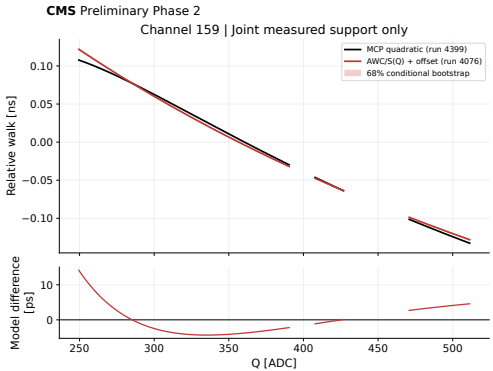
Channel 154: a second shape-comparison example



The same objective, restricted to the measured overlap, determines AWC and the offset for channel 154. Both plotted curves are calibrated models, not matched events from different runs. The lower panel shows the remaining shape mismatch, with no additional line subtracted. This measures model-shape closure, not corrected-event timing resolution.

Conditional bootstrap band: 4076 slope replicas, refitted AWC and offset, fixed 4399 quadratic.

Channel 159: missing support is not extrapolation

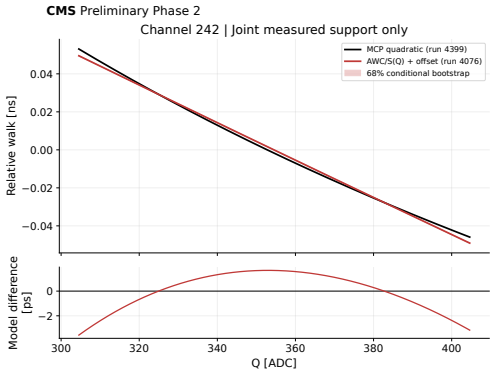


Only intervals with populated profile bins in both runs contribute to the alignment. Breaks in the displayed curves mark excluded charge intervals. They do not represent discontinuities in either analytic model.

The lower panel uses the same supported points. All displayed intervals participate in the common AWC-and-offset fit. It is a model difference, not an event-time distribution.

The comparison excludes 703 of the 3000 initial grid points for this channel. No line is drawn through those gaps.

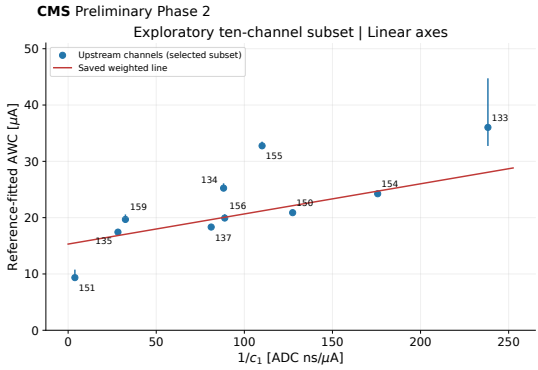
Downstream example: channel 242



This example applies the method to a downstream channel. In run 4399, upstream bar 8 supplies the coincidence pair alongside the downstream target pair and MCP indicator.
In run 4076, the target downstream pair is required, without fixing an upstream pair. The curves use only their common measured support. Downstream comparisons also assume the same nominal detector setup in both runs. Selection-dependent response differences remain unquantified.

The lower panel is a model difference, not a timing-resolution measurement. The other module is not used as an internal timing reference to determine AWC here.

A selected subset does not establish a universal law



The ten-channel upstream subset previously selected for discussion is shown on linear axes. Bars are slope-only statistical AWC bootstrap intervals; channel numbers identify the points.

The weighted straight-line fit has $\chi^2/\mathrm{ndf} = 538.6/8$ under its limited error model. It does not adequately describe the variation. The subset was selected interactively and is illustrative, not quality-selected or an unbiased population-level validation.

The fit treats x as fixed and uses half the 16–84% AWC span as its symmetric uncertainty. Correlations and reference errors are omitted.

Channel coverage and limitations

The explored set contains 14 upstream and 8 downstream channels, with 21 completed alignments. Only channel 238 lacks measured overlap. No additional quality cut is applied to this summary.

Channels	Observed limitation and treatment
238	No common measured charge range; no AWC estimate.
235, 244	Overlap spans only 3.10 and 16.99 ADC, respectively; one profile mean from each run lies within each overlap. Fits retained.
130, 236, 239	Slope-only AWC intervals include zero and span hundreds of μA. Fits retained as poorly constrained cases.
136, 245	Negative central AWC values are retained, not accepted as validated positive-amplitude calibrations.

The ten-channel illustration is unchanged: 133, 134, 135, 137, 150, 151, 154, 155, 156 and 159. This subset does not define a channel-selection criterion.

Table entries are diagnostic examples, not additional exclusions. Small curve mismatch over narrow support does not validate a calibration. The full per-channel support and interval table is saved with the figure data.

Uncertainties

Displayed quantity	Meaning
Profile bars	SD of selected hits in each charge bin; not the uncertainty on the mean.
Fit weights	SEM of each profile mean; not the SD.
Slope band	16–84% range from 1000 bootstrap linear fits with fixed bin edges.
AWC bars / walk band	Propagate the slope replicas through a new AWC-plus-offset alignment to the same fixed reference curve.

These definitions refer to the original displayed bands and bars. The next table additionally propagates the run-4399 quadratic covariance. Both prescriptions exclude uncertainty in response transfer between runs, clock and fine-time calibration, selection and model choices, and cross-channel correlations. Neither is a total calibration uncertainty or a timing-resolution estimate.

MCP-fit uncertainty and charge weighting

For each slope-bootstrap replica, propagate the correlated MCP coefficient covariance through the AWC-plus-offset fit. The Gaussian mixture gives the 16–84% interval; central AWC values and comparison curves are unchanged.

All table values are in μA. "Both runs" adds MCP-fit statistics, not cross-run or model systematics. The original plots retain slope-only statistical bands.

Channel	AWC	Slope only	Both runs	Max. weight shift
133	36.03	[32.73, 44.75]	[32.48, 45.72]	4.11
154	24.25	[23.92, 24.86]	[22.98, 25.84]	0.85
137	18.33	[18.12, 18.61]	[17.91, 18.80]	2.05
150	20.89	[20.65, 21.15]	[20.41, 21.38]	1.13

Maximum weighting shift: largest $|\Delta\mathrm{AWC}|$ using uniform $1/Q$ or either run's charge population instead of uniform Q. The overlap and free offset are unchanged. These shifts are not statistical errors and are not added in quadrature.

All 21 overlap channels were checked; channel 238 has no overlap. A shared reference-time offset affects only the fitted offset, not AWC. These are conditional statistical intervals, with weighting sensitivity reported separately, not total calibration uncertainties.

Result and next validation step

- A dual-threshold measurement provides a charge-dependent slope proxy; an inverse-slope function can be aligned to independent MCP walk curves.
- Restricting alignment to jointly measured charge support avoids determining AWC from extrapolation.
- Reference-calibrated AWC varies between channels. A straight line against $1/c_1$ is insufficient for the explored subset.
- The present calibration still uses the MCP to determine AWC. It does not establish reference-free AWC extraction by itself.
- Before use without MCP, an internal predictor must be tested on held-out bars and configurations, including uncertainties in transfer and in the target AWC values.

The following backup explores such a predictor; it is not a final performance claim or a demonstration with cosmic rays.

Backup: an internal predictor for channel AWC

For $W = A/S$ and $S = c_0 + c_1 Q$,

$$-\frac{dW}{d\ln Q} = \frac{A\, c_1 Q}{S(Q)^2}.$$

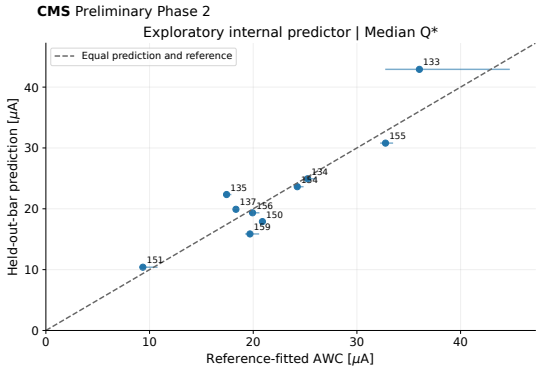
If this local walk derivative is approximately shared between channels at a prescribed typical charge Q_*, a candidate model is

$$A_{\mathrm{pred}} = K\, \frac{S(Q_*)^2}{c_1 Q_*}.$$

Here Q_* is the median charge of the run-4076 quality- and peak-right-selected sample. It is computed once per channel; A_{pred} remains constant when correcting individual events.

The common K is learned from MCP-calibrated training channels. Predictions for new channels use internal measurements, but model development uses the reference. The shared derivative and choice of Q_* remain hypotheses requiring independent validation across operating conditions, without tuning to the test results.

Backup: predict a bar excluded from the model fit



Each point compares reference-fitted AWC with a prediction made after excluding that channel's entire bar from the fit of K. Ten upstream channels span six held-out bar groups.
The RMS prediction error is 3.22 μA; the RMS fractional error is 14.1%.
Horizontal intervals are conditional reference-AWC intervals, not predictive intervals. This small selected sample leaves input–target correlations and full prediction uncertainties unpropagated. It does not demonstrate reference-free calibration.

Exploratory median-Q_* candidate. The dashed line indicates equality, not a fitted result. A general 10% precision claim is not established.

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[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[5] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

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[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) `https://doi.org/10.1016/0168-9002(94)01239-3`.

[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured- ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[12] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.